# Assessing the Accuracy of Multiple Sequence Alignments, when Analysing Protein

Moya Caulfield
BGG, School of Biological and Chemical Sciences, University of Galway

## 1. Summary

Multiple sequence alignment (MSA) is an essential bioinformatics tool used to identify regions of homology and detect variation in biological sequences. It has a wide range of applications in the life sciences, including protein function and structure prediction, phylogenetics, and genomics. MSAs are inferred and errors are common, and this can have an impact on downstream analyses.

In this project, the aim is to inform researchers about which MSA method to choose, by ranking selected MSA methods. To do this, we focus on alignment error in datasets that is composed of highly similar sequences, specifically alternatively spliced gene products. I evaluate each method by how closely an inferred MSA matches a reference MSA. The MSA methods chosen were ClustalW, Clustal Omega, MUSCLE, MAFFT (G-INS-i), ProbCons, and T-Coffee. Varying gap patterns and gap lengths were explored.

I found that gap patterns bias alignment more than gap lengths. Iterative MSA methods show improved handling of gaps, and progressive MSA methods show greater sensitivity to gap placement.

**Aim:** To explore the possibility of ranking popular Multiple Sequence Alignment methods to guide researchers in selecting the most suitable and accurate algorithms.

## 2. Materials and Methods

1. Make 4 reference alignments (with gaps of 10/30 residues, gap patterns A/B) (Fig 1.)
2. Delete gaps in reference alignments
3. Use MSA methods to infer alignments (i.e., ClustalW, Clustal Omega, etc)
4. Use the Total Column (TC) Score distance to compare the inferred MSA (Fig. 2) to the reference MSA (Fig. 1)

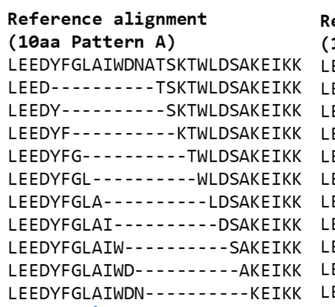


Fig. 1a – Reference alignment for tiled, overlapping gaps of 10 residues (a similar reference alignment was created for gaps of 30 residues).

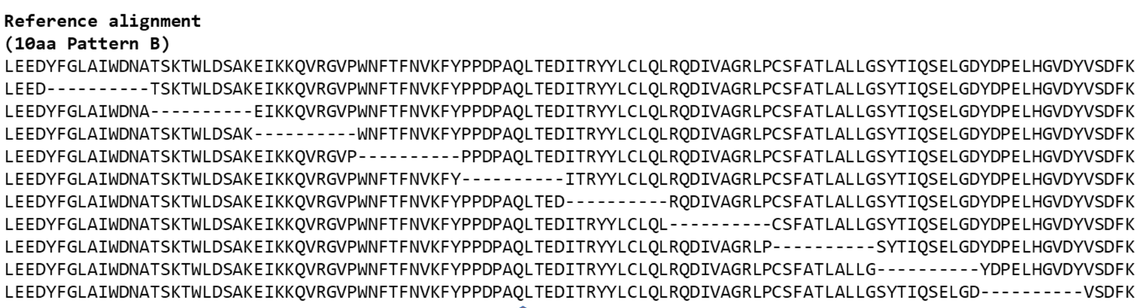


Fig. 1b – Reference alignment for tiled, non-overlapping gaps of 10 residues (a similar reference alignment was created for gaps of 30 residues).

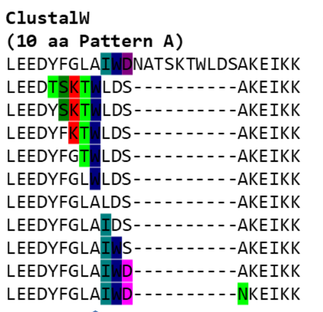


Fig. 2a – Inferred alignment made using the MSA method ClustalW for overlapping gaps of 10 amino acids.

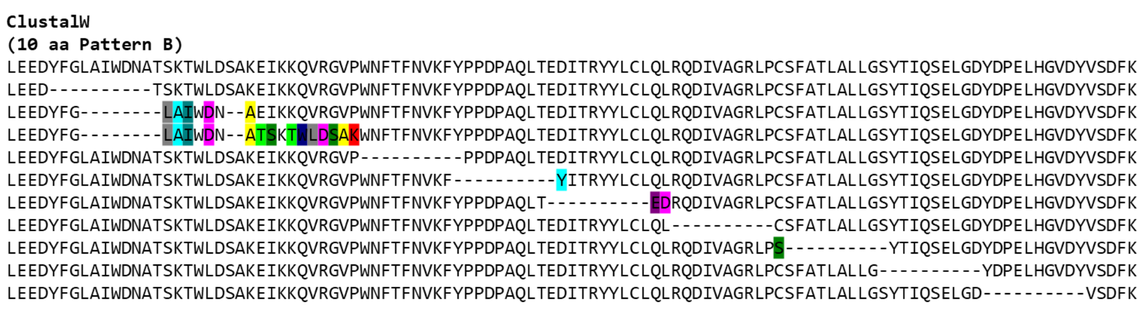


Fig. 2b – Inferred alignment using the MSA method ClustalW for non-overlapping gaps of 10 amino acids.

### MSA programs encounter difficulties with non-overlapping gaps

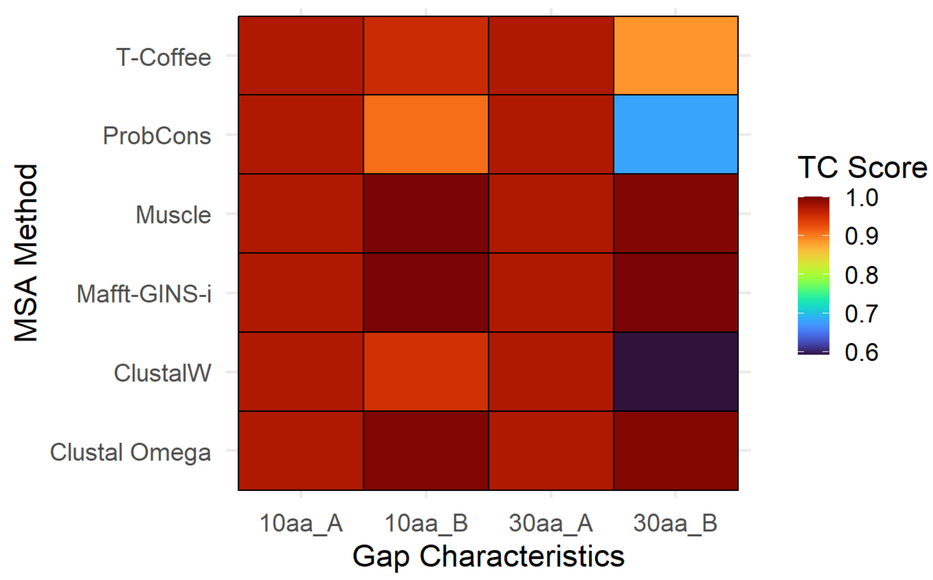


Fig. 3 – TC Scores for overlapping gaps (pattern A), either 10 or 30 residues removed, remain the same but for non-overlapping gaps (pattern B), the accuracy differs significantly.

### Iterative methods handle gaps more effectively than progressive or probabilistic methods

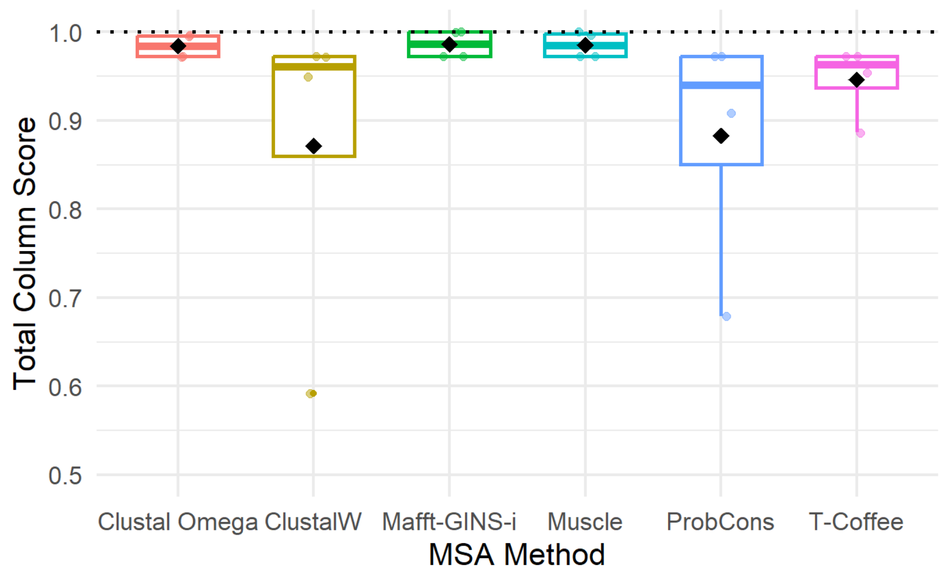


Fig. 4 – Iterative methods (MUSCLE, MAFFT (G-INS-i)) are the only methods to achieve an inferred MSA identical to the reference alignment (TC Score=1). ClustalW and ProbCons obtained a wide range of TC Scores with significant outliers.

### An Increase in Gap Length Reduces Alignment Accuracy in Most MSA Methods

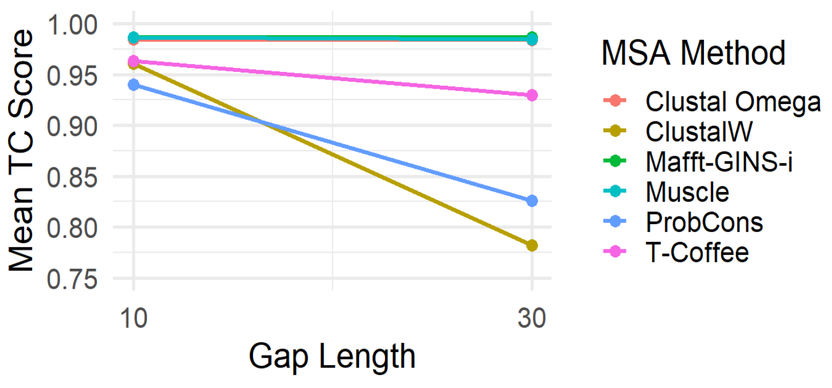


Fig. 5 – Mean TC scores for MAFFT (G-INS-i), MUSCLE, and Clustal Omega remain unchanged across gap lengths of 10 and 30 amino acids. T-Coffee shows a slight decrease in accuracy with larger gaps, while ProbCons and ClustalW decline significantly. The gap length of 10 uses the mean TC score from overlapping and non-overlapping sequences with 10 amino acids removed.

### Non-overlapping Gaps can either Improve or Decrease the Accuracy of Alignments

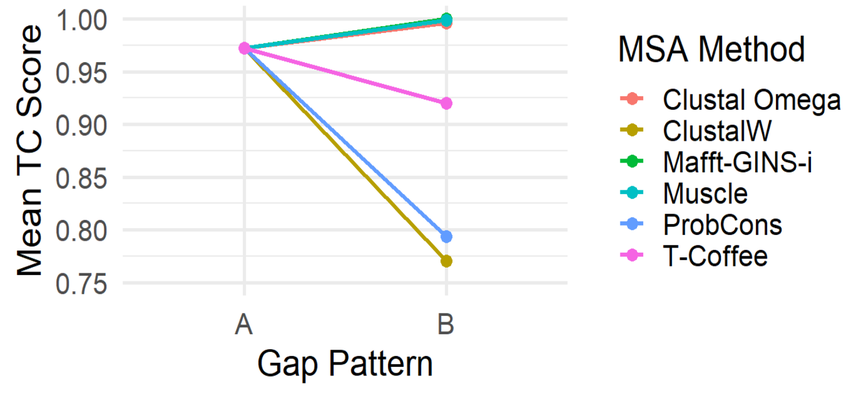


Fig. 6 – Mean TC Scores for MAFFT (G-INS-i), MUSCLE, and Clustal Omega increase with non-overlapping compared to overlapping gaps. T-Coffee shows a slight decrease in accuracy in non-overlapping gaps, while ProbCons and ClustalW both show a significant decrease when gaps are non-overlapping. The gap pattern A uses the mean TC Score from sequences with 10 and 30 amino acids removed that are overlapping.

## 3. Results and Discussion

- Results are presented in Fig. 3-6.
- Overall, MUSCLE and MAFFT (G-INS-i) are the most accurate MSA methods tested – we can conclude that iterative methods provide the most reliable and accurate alignments.
- Clustal Omega performs reasonably well but is consistently slightly less accurate than MUSCLE and MAFFT (G-INS-I).
- T-Coffee and ProbCons show intermediate performance, with good accuracy under simple gap conditions but reduced robustness when gap patterns become more complex.
- ClustalW is the least accurate method tested across varying gap conditions – this is notable as it is an incredibly popular MSA method.
- Choosing an appropriate MSA tool is critical for maintaining alignment accuracy, especially in sequences with large or complex gaps i.e. alternatively spliced gene products.

### Acknowledgements

I gratefully acknowledge my supervisor, Professor Lars Jermiin for his guidance and support throughout this project.